

Camel urine inhibits the cytochrome P450 1a1 gene expression through an AhR-dependent mechanism in Hepa 1c1c7 cell line.

AIM OF THE STUDY: Drinking camel urine has been used traditionally to treat numerous cases of cancer yet, the exact mechanism was not investigated. Therefore, we examined the ability of three different camel urines (virgin, lactating, and pregnant source) to modulate a well-known cancer-activating enzyme, the cytochrome P450 1a1 (Cyp1a1) in murine hepatoma Hepa 1c1c7 cell line. **MATERIALS AND METHODS:** The effect of different camel urines, compared to bovine urines, on Cyp1a1 mRNA was determined using real-time polymerase chain reaction. Cyp1a1 protein and catalytic activity levels were determined using Western blot analysis and 7-ethoxyresorufin as a substrate, respectively. The role of aryl hydrocarbon receptor (AhR)-dependent mechanism was determined using electrophoretic mobility shift assay (EMSA) and the AhR-dependent luciferase reporter gene. **RESULTS:** All types of camel, but not bovine, urines differentially inhibited the induction of Cyp1a1 gene expression by TCDD, the most potent Cyp1a1 inducer and known carcinogenic chemical. Importantly, virgin camel urine showed the highest degree of inhibition at the activity level, followed by lactating and pregnant camel urines. Furthermore, we have shown that virgin camel urine significantly inhibited the TCDD-mediated induction of Cyp1a1 at the mRNA and protein expression levels. Mechanistically, the ability of virgin camel urine to inhibit Cyp1a1 was strongly correlated with its ability to inhibit AhR-dependent luciferase activity and DNA binding as determined by EMSA, suggesting that AhR-dependent mechanism is involved. **CONCLUSIONS:** The present work provides the first evidence that camel urine but not that of bovine inhibits the TCDD-mediated toxic effect by inhibiting the expression of Cyp1a1, at both transcriptional and post-transcriptional levels through an AhR-dependent mechanism. Copyright © 2010 Elsevier Ireland Ltd. All rights reserved.